

3.2 Development Stack and Technical Framework

The following point-form list summarizes the technical framework. Each entry states the tool or library and briefly describes how it contributes to the implementation.

- **Figma:** Leveraged for high-fidelity prototyping and UI/UX design. This tool allowed for the iterative testing of accessibility features, such as color contrast and element scaling, before the commencement of the development phase.
- **Flutter (Dart):** Employed as the primary cross-platform framework to ensure seamless deployment on both Android and iOS from a single codebase. The Flutter rendering stack supports responsive, accessible, and custom UI components appropriate for low-vision interaction design.
- **Google ML Kit (Digital Ink Recognition):** The `google_mlkit_digital_ink_recognition` plugin connects the handwriting canvas to on-device digital-ink models provided by Google ML Kit. Ink traces are interpreted locally so practice feedback can be generated without transmitting raw strokes to a remote backend.
- **Flutter APIs and flutter_tts:** Utilized to integrate Text-to-Speech (TTS) capabilities. This supports real-time, voice-guided instruction and uses the host platform speech engine, which aligns with native auditory feedback and screen-reader ecosystems.
- **Bilingual interface and localization approach:** English and Chinese interface strings, together with Cantonese and English voice output, are coordinated through Dart application state and saved preferences. Dynamic language switching is therefore handled in application logic rather than through `intl` or `flutter_localizations` dependencies.
- **shared_preferences, Semantics, and flutter_lints:** The `shared_preferences` plugin persists UI and voice language selections across sessions. `Semantics` metadata augments widgets for accessibility services, and `flutter_lints` enforces recommended static-analysis rules for maintainable Dart source.